

ances). The 12V supply to drive the relays is also provided through the main circuit (Fig. 2).

Software

The software program (home.c) for the microcontroller is written in 'C' language and compiled using Keil software to generate the hex code. The microcontroller receives control commands as ASCII values sent from a laptop or Android mobile, and controls the corresponding devices as per Table I.

BlueTerm. BlueTerm is an Android application that is a VT-100 terminal emulator for communicating with any serial device using a Bluetooth serial adaptor. One can download it from 'Android Apps on Google Play' and the link is: <https://play.google.com/store/apps/details?id=es.pymasde.bluetooth&hl=en>

Install the application on your Android phone and follow the steps for operating relays:

1. Start the Bluetooth connection of your mobile phone or tablet.
2. Open the 'BlueTerm' application on your mobile or tablet.
3. Select 'Connect device' from the 'Option' menu after scanning 'New devices.'
4. 'Bluetooth Serial Module' will be available under the 'Select a device to connect' list.
5. Pair 'Bluetooth Serial Module' by providing pair code '1234'.
6. Start sending ASCII characters as shown in Table I; you will see different relays going on and off corresponding to the values sent.

Bluetooth application software. Bluetooth application software is developed using Visual Basic 2008 Express Edition, which is part of Microsoft Visual Studio Express. Microsoft Visual Studio Express is a set of freeware Integrated Development Environments (IDE) developed by Microsoft that are lightweight versions of the Microsoft Visual Studio product line. The idea of Express editions is to provide streamlined, easy-to-use and easy-to-learn IDEs for users other than

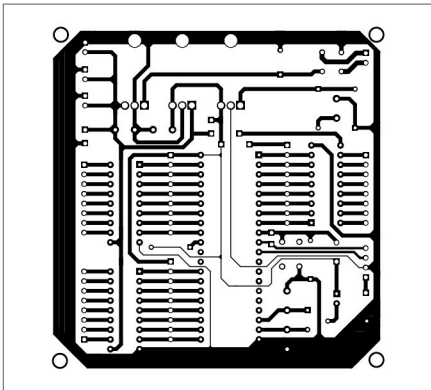


Fig. 7: An actual-size, single-side PCB layout for the 'Home Automation through Bluetooth' application

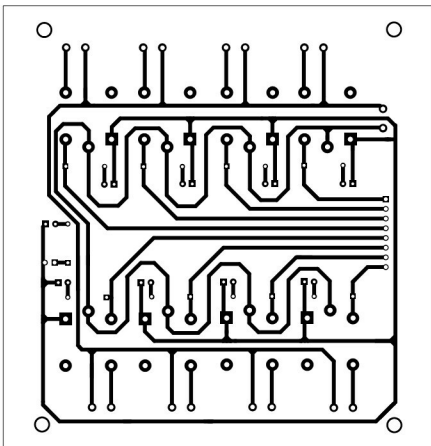


Fig. 8: An actual-size single-side PCB layout for the relay board